

# NCFED on the Wire

A packet-level walkthrough of NetClaw N2N federation — peering **as65007-7.7.7.7 (Nick)** ⇄ **as65001-4.4.4.4 (John / johns-risk)**, including a complete skill-delegation lifecycle with live authorization denials.

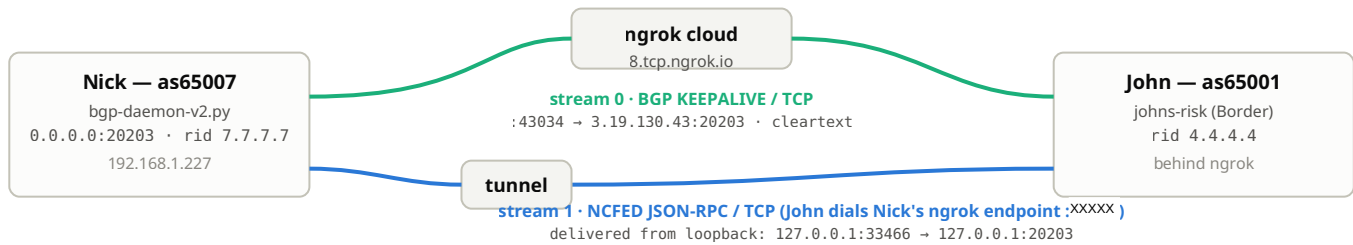
captures/n2n-session-20260714.pcap · 2026-07-14 14:03:05 → 14:30:21 EDT · capture filter tcp port 20203 · interface any (linux-sll2) · decoded with an NCFED frame parser implementing the bgp/federation/channel.py wire format · frame numbers reference the pcap

|                  |                       |                     |                                     |                                   |                                |
|------------------|-----------------------|---------------------|-------------------------------------|-----------------------------------|--------------------------------|
| 1,059<br>packets | 27 m 15 s<br>duration | 115 kB<br>wire data | 446<br>protocol messages<br>decoded | 137<br>JSON-RPC req/resp<br>pairs | 0<br>retransmits / dup<br>ACKs |
|------------------|-----------------------|---------------------|-------------------------------------|-----------------------------------|--------------------------------|

## Executive summary

This capture records a complete **eN2N skill-delegation lifecycle** between two federated NetClaw agents. John's agent opened chat sessions, attempted to delegate the pyats-health-check skill twice and was **denied both times** (JSON-RPC error -32001 in ~20 ms), the authorization layer failing closed exactly as designed. After the operator granted the skill at 14:09:05 — a local API action, deliberately invisible on the wire — identical re-submissions were **accepted in 6 ms**. One task completed in 10 m 25 s and returned a 3,114-byte fleet health report; the other **stalled in “working”** and refused a cancel — the one genuine anomaly. Transport was flawless: zero retransmissions, zero duplicate ACKs, heartbeats never missed.

## Topology — what the two TCP streams actually are



Both streams are the **same peering** seen from different sides. Stream 0 is Nick's daemon dialing John's published ngrok TCP endpoint — a genuine RFC 4271 BGP session used for mesh liveness (KEEPALIVES only in this window; no UPDATES). Stream 1 is John dialing Nick's ngrok endpoint; the local ngrok agent terminates the tunnel and delivers it from loopback, which is why the NCFED channel appears as 127.0.0.1 ⇄ 127.0.0.1. Peer Byrn (as65099) has no direct endpoint and generated no traffic in the window.

## Wire formats observed

|                                           |                           |                     |  |
|-------------------------------------------|---------------------------|---------------------|--|
| <b>NCFED frame header — every message</b> |                           | 5 bytes, big-endian |  |
| <b>length</b><br>uint32 !! payload bytes  | <b>flags</b><br>bit0=cont |                     |  |
| 0                                         | 4                         | 5                   |  |

|                                        |                      |         |  |
|----------------------------------------|----------------------|---------|--|
| <b>NCFED heartbeat — 30 s liveness</b> |                      | 5 bytes |  |
| <b>length = 0</b><br>00 00 00 00       | <b>flags 0</b><br>00 |         |  |
| 0                                      | 4                    | 5       |  |

|                                              |                               |                                 |          |  |
|----------------------------------------------|-------------------------------|---------------------------------|----------|--|
| <b>NCFED handshake — once per connection</b> |                               |                                 | 13 bytes |  |
| <b>magic "NCFED"</b><br>4e 43 46 45 44       | <b>AS number</b><br>uint32 !! | <b>router-id</b><br>IPv4 packed |          |  |
| 0                                            | 5                             | 9                               | 13       |  |

|                                                |  |                                  |          |  |
|------------------------------------------------|--|----------------------------------|----------|--|
| <b>BGP KEEPALIVE (RFC 4271) — mesh session</b> |  |                                  | 19 bytes |  |
| <b>marker</b><br>16 × 0xFF                     |  | <b>length type</b><br>0x0013 = 1 | <b>4</b> |  |
| 0                                              |  | 16                               | 1819     |  |

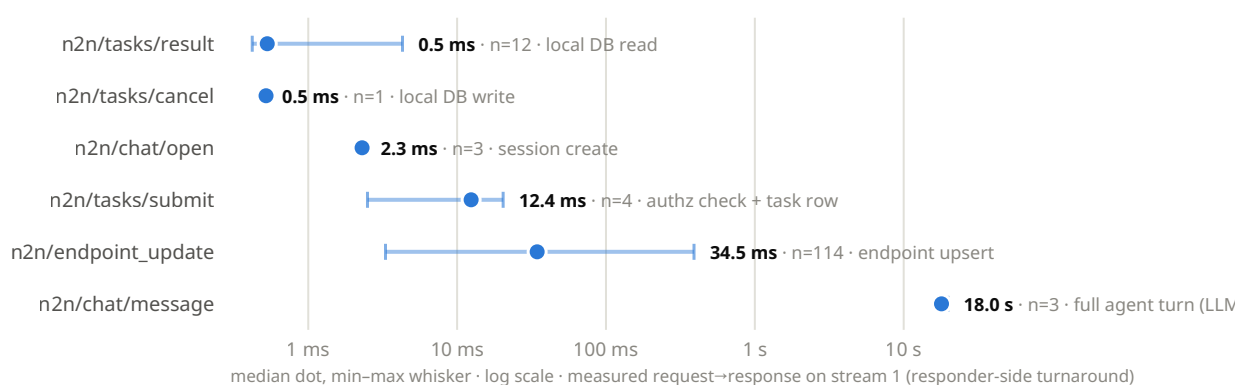
NCFED is length-prefixed JSON-RPC 2.0 over TCP: every message is a 5-byte !IB header followed by a UTF-8 JSON body. A zero-length frame is a heartbeat. Payloads over 64 kB are chunked with the continuation flag (none needed in this session — largest frame was 3,114 B). The 13-byte handshake exchanges AS number and router-id before any frames flow; this session's handshakes pre-date the capture (see ncfed-session.pcap from 2026-07-13 for a captured handshake, n2n/hello, and the 46 kB inventory exchange).

## JSON-RPC method inventory — everything seen on stream 1

| Method              | Direction   | Count | Avg req | Avg resp | Median RTT | Purpose (as observed)                                                       |
|---------------------|-------------|-------|---------|----------|------------|-----------------------------------------------------------------------------|
| n2n/endpoint_update | both → both | 114   | 153 B   | 75 B     | 34.5 ms    | re-announce own public endpoint (ngrok host:port); ran every ~28 s per side |
| n2n/chat/open       | John → Nick | 3     | 110 B   | 127 B    | 2.3 ms     | create operator chat session (returns session_id)                           |
| n2n/chat/message    | John → Nick | 3     | 303 B   | 308 B    | 18.0 s     | chat turn — response carries the full agent reply (latency = LLM turn)      |
| n2n/tasks/submit    | John → Nick | 4     | 388 B   | 137 B    | 12.4 ms    | async skill delegation; authz check → task_id or error -32001               |
| n2n/tasks/result    | John → Nick | 12    | 136 B   | 389 B    | 0.5 ms     | poll task state; final poll returned the 3,114 B completed report           |
| n2n/tasks/cancel    | John → Nick | 1     | 137 B   | 127 B    | 0.5 ms     | cancel request — returned cancelled=false (finding 2)                       |
| heartbeat (len=0)   | both → both | 115   | 5 B     | —        | —          | liveness proof; 30.0 s server-side, ~27.3 s client-side                     |

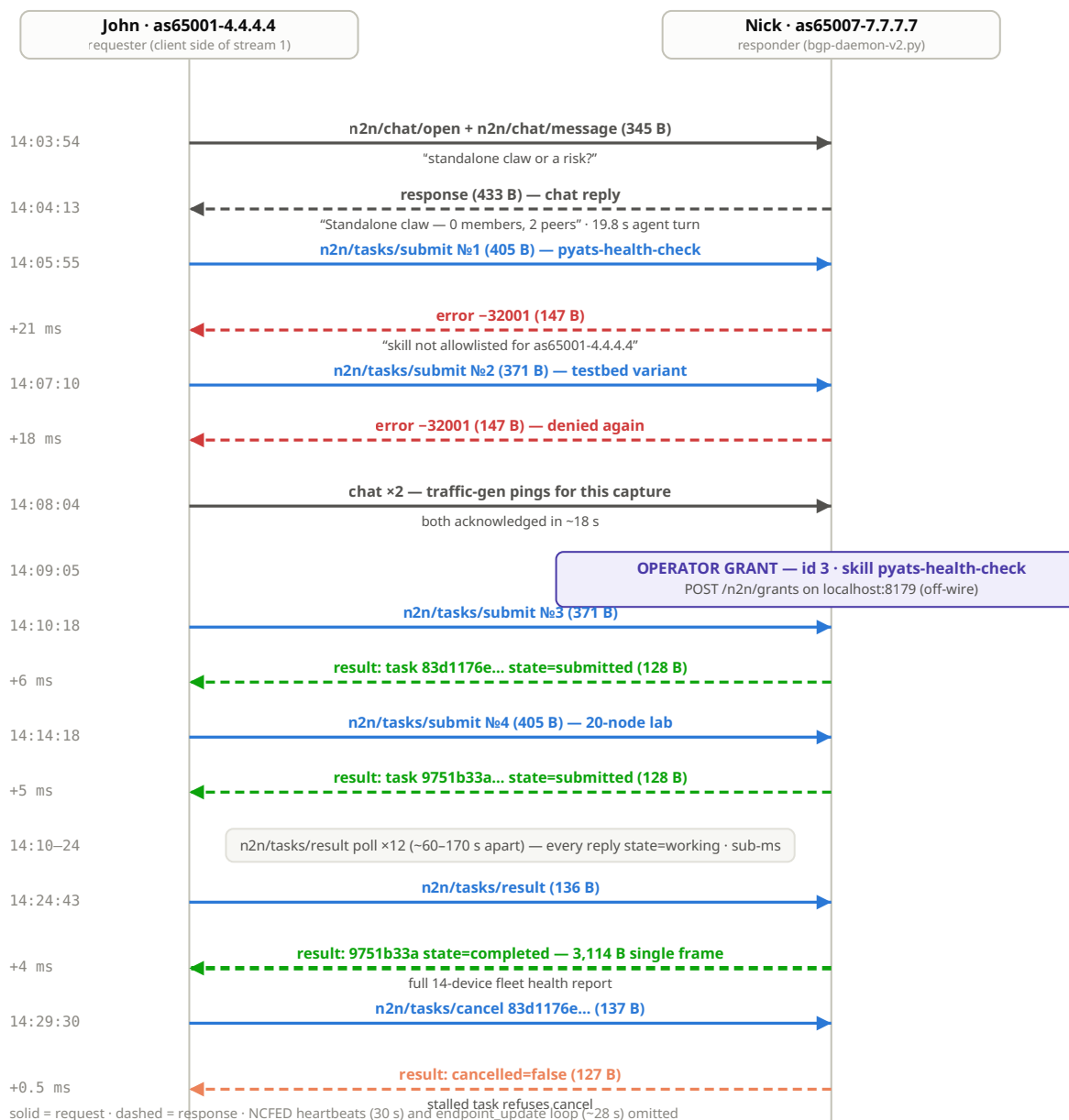
Request IDs are namespaced by sender identity ("as65001-4.4.4.4:35") — both sides can issue requests on one connection without ID collisions. Every request got exactly one response; no orphans, no duplicates, no reordering.

## Response latency by method — three timing regimes

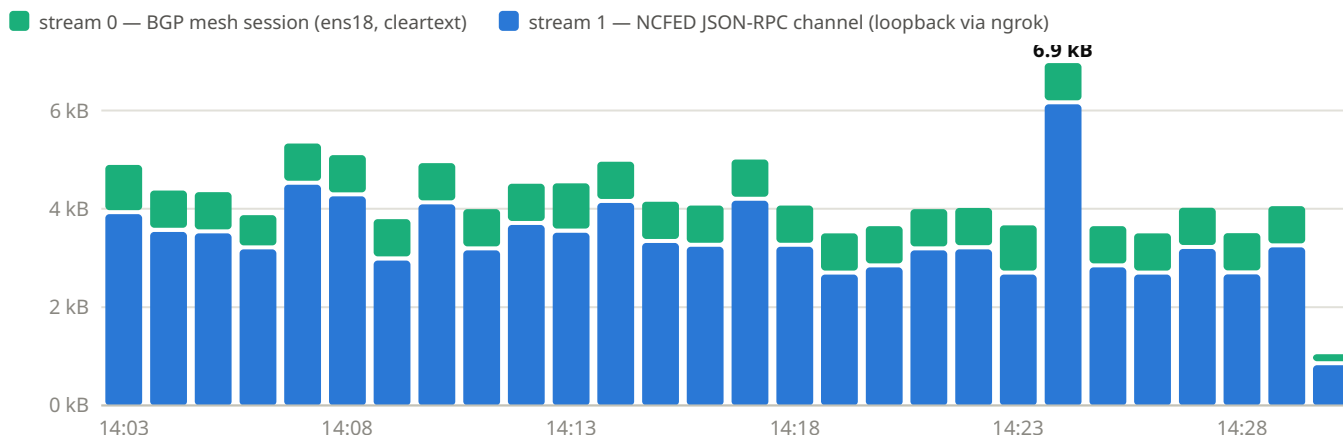


The spread is diagnostic: **sub-millisecond** methods are pure SQLite reads/writes in the daemon; **tens of milliseconds** involve authz checks or endpoint upserts; **~18 seconds** is a full LLM agent turn — chat/message blocks synchronously on the reply, which is exactly why tasks/submit is async (submit → poll → result) for long-running skills.

## Session sequence — the delegation lifecycle

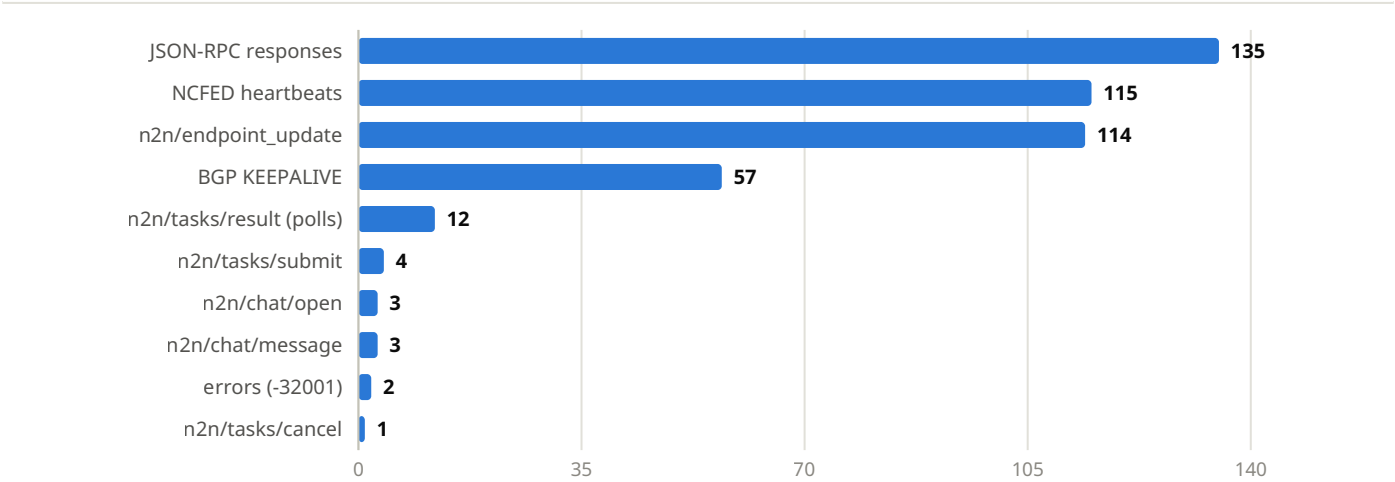


## Traffic over time — bytes per minute



The ≈3.4 kB/min floor is pure control plane: heartbeats, BGP keepalives, and the endpoint re-announce loop (finding 3). The 14:24 spike is the completed health-check result.

Message-type breakdown — all 446 decoded messages



Substantive agent work — chat, submits, polls, one result, one cancel — is 25 messages (5.6%). Everything else is liveness and endpoint chatter.

Liveness cadence & transport health

| Timer                           | Observed | Jitter (min-max) | TCP pathology check (tshark analysis flags) | Count |
|---------------------------------|----------|------------------|---------------------------------------------|-------|
| NCFED heartbeat — Nick's daemon | 30.0 s   | 30.0 – 30.0 s    | Retransmissions                             | 0     |
| NCFED heartbeat — John's client | 27.3 s   | 27.2 – 29.8 s    | Duplicate ACKs                              | 0     |
| BGP KEEPALIVE — Nick (local)    | 60.0 s   | 60.0 – 60.0 s    | Zero-window events                          | 0     |
| BGP KEEPALIVE — John (remote)   | 54.5 s   | 54.5 – 58.4 s    | Out-of-order segments                       | 0     |

The daemon's timers are metronomic (jitter only on the remote/client sides, where event-loop scheduling shows). The 3,114 B result crossed loopback in a **single TCP segment** (64 kB loopback MSS); on the internet path the same frame would span ~3 segments at a 1,460 B MSS — still one NCFED frame after reassembly.

Session timeline — the conversation, diagnosed

| Time (EDT)  | Direction   | Verdict     | Event                                                                                                                                                                                        |
|-------------|-------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 14:03:05    | —           | · INFO      | <b>Capture starts</b><br>tcpdump -i any 'tcp port 20203'; both channels already established (mid-stream — handshake & inventory pre-date the window).                                        |
| 14:03:54    | John → Nick | » CHAT      | <b>n2n/chat/open + message</b><br>“Is your NetClaw a standalone claw or a risk?” — federation topology question.                                                                             |
| 14:04:13    | Nick → John | » CHAT      | <b>chat reply (433 B)</b><br>“Standalone claw — 0 members, 2 federated peers.” 19.8 s round-trip (agent turn).                                                                               |
| 14:05:55    | John → Nick | × DENIED    | <b>n2n/tasks/submit №1 · frame 116</b><br>pyats-health-check vs 20-Node v2 lab. <b>DENIED</b> in 21 ms — error -32001, frame 118.                                                            |
| 14:07:10    | John → Nick | × DENIED    | <b>n2n/tasks/submit №2</b><br>Testbed-wide variant. <b>DENIED</b> — -32001 again. Fail-closed authz holds.                                                                                   |
| 14:08:04    | John → Nick | » CHAT      | <b>chat x2 + replies</b><br>Traffic-gen pings for this capture demo; both round-trips acknowledged (~18 s each).                                                                             |
| 14:09:05    | operator    | ● GRANT     | <b>grant applied (off-wire)</b><br>POST /n2n/grants on localhost:8179 → grant id 3, skill pyats-health-check for as65001-4.4.4.4. Correlated from federation.db — not visible on port 20203. |
| 14:10:18    | John → Nick | ✓ ACCEPTED  | <b>n2n/tasks/submit №3 · frame 295</b><br><b>ACCEPTED</b> in 6 ms → task 83d1176e... state=submitted (frame 296). First post-grant delegation.                                               |
| 14:14:18    | John → Nick | ✓ ACCEPTED  | <b>n2n/tasks/submit №4</b><br>20-node lab variant → task 9751b33a... Both tasks now in flight.                                                                                               |
| 14:10–14:24 | John → Nick | · INFO      | <b>n2n/tasks/result x12</b><br>Poll loop across both tasks, ~60–170 s apart; every reply state=working; each answered in <1 ms.                                                              |
| 14:24:43    | Nick → John | ✓ COMPLETED | <b>task 9751b33a completed · frame 853</b><br>3,114 B result in a single NCFED frame (one loopback TCP segment) — full 14-device fleet health report. 10 m 25 s wall-clock.                  |

|          |             |                      |                                                                                                                              |
|----------|-------------|----------------------|------------------------------------------------------------------------------------------------------------------------------|
| 14:29:30 | John → Nick | <div>△ ANOMALY</div> | <b>n2n/tasks/cancel 83d1176e</b><br>Reply: cancelled=false in 0.5 ms. Task №3 never left “working” and refused cancellation. |
| 14:30:21 | —           | <div>· INFO</div>    | <b>Capture stops</b><br>1,059 packets on file.                                                                               |

## Packet dissections — real captured bytes, annotated

### Frame 1 — BGP KEEPALIVE from John's endpoint (full stack)

☐ linux-sll2 (20 B) ☒ IPv4 (20 B) ☒ TCP + opts (32 B) ☐ BGP marker/len ☐ BGP fields

0000 08 00 00 00 00 00 00 02 00 01 00 06 bc 9a 8e eb .....  
0010 f1 41 00 00 45 00 00 47 c9 cf 40 00 73 06 f6 17 ·A·E·G·@·s·  
0020 03 13 82 2b c0 a8 01 e3 4e eb a8 1a 06 f2 e5 34 ··+·N·4  
0030 8c 2c 33 af 80 18 00 f5 83 9b 00 00 01 01 08 0a ·,3·  
0040 f8 da 4c d8 5f 58 5e 21 ff ff ff ff ff ff ff ··L·\_X^!·  
0050 ff ff ff ff ff ff ff ff 00 13 04 .....

14:03:05.610276 · 3.19.130.43:20203 → 192.168.1.227:43034 · [PSH,ACK] — RFC 4271 §4.4: 16-byte all-ones marker, length 0x0013 (19), type 0x04 (KEEPALIVE). TTL 0x73 (115) — the path transits ngrok's edge. This is the entire mesh-liveness message; no BGP UPDATE appeared all session.

### Frame 5 — NCFED heartbeat (the whole message is 5 bytes)

☐ NCFED header — length (uint32) + flags ☐ JSON-RPC 2.0 payload (UTF-8)

0000 00 00 00 00 00 .....

14:03:12.588192 · 127.0.0.1:33466 → 127.0.0.1:20203 — length=0, flags=0. Any inbound frame proves liveness; NCFED\_HEARTBEAT\_MISS\_LIMIT consecutive silent intervals close the channel.

### Frame 7 — n2n/endpoint\_update (the re-announce loop, finding 3)

☐ NCFED header — length (uint32) + flags ☐ JSON-RPC 2.0 payload (UTF-8)

0000 00 00 00 94 00 7b 22 6a 73 6f 6e 72 70 63 22 3a .....{"jsonrpc":  
0010 22 32 2e 30 22 2c 22 69 64 22 3a 22 61 73 36 35 "2.0","id":"as65  
0020 30 30 37 2d 37 2e 37 2e 37 2e 37 3a 32 33 22 2c 007-7.7.7.7:23",  
0030 22 6d 65 74 68 6f 64 22 3a 22 6e 32 6e 2f 65 6e "method":"n2n/en  
0040 64 70 6f 69 6e 74 5f 75 70 64 61 74 65 22 2c 22 dpoint\_update", "  
0050 70 61 72 61 6d 73 22 3a 7b 22 69 64 65 6e 74 69 params":{"identi  
0060 74 79 22 3a 22 61 73 36 35 30 30 37 2d 37 2e 37 ty":"as65007-7.7  
0070 2e 37 2e 37 22 2c 22 65 6e 64 70 6f 69 6e 74 22 .7.7","endpoint"  
0080 3a 22 38 2e 74 63 70 2e 6e 67 72 6f 6b 2e 69 6f : "8.tcp.ngrok.io  
0090 3a 31 30 34 31 36 22 7d 7d :xxxxx "}}

14:03:12.762144 · Nick → John — header 00 00 00 94 00: length 0x94 = 148 bytes, flags 0. Nick re-announces his own tunnel endpoint (8.tcp.ngrok.io:xxxxx ). This exact message repeated 57× per side with an unchanged endpoint value.

### Frame 118 — authorization denial (the -32001 error, complete)

☐ NCFED header — length (uint32) + flags ☐ JSON-RPC 2.0 payload (UTF-8)

0000 00 00 00 8e 00 7b 22 6a 73 6f 6e 72 70 63 22 3a .....{"jsonrpc":  
0010 22 32 2e 30 22 2c 22 69 64 22 3a 22 61 73 36 35 "2.0","id":"as65  
0020 30 30 31 2d 34 2e 34 2e 34 2e 34 3a 33 35 22 2c 001-4.4.4.4:35",  
0030 22 65 72 72 6f 72 22 3a 7b 22 63 6f 64 65 22 3a "error":{"code":  
0040 2d 33 32 30 30 31 2c 22 6d 65 73 73 61 67 65 22 -32001,"message"  
0050 3a 22 73 6b 69 6c 6c 20 27 70 79 61 74 73 2d 68 : "skill 'pyats-h  
0060 65 61 6c 74 68 2d 63 68 65 63 6b 27 20 6e 6f 74 ealth-check' not  
0070 20 61 6c 6c 6f 77 6c 69 73 74 65 64 20 66 6f 72 allowlisted for  
0080 20 61 73 36 35 30 30 31 2d 34 2e 34 2e 34 2e 34 as65001-4.4.4.4  
0090 22 7d 7d "}}

14:05:55.987033 · Nick → John · 21 ms after submit frame 116 — header length 0x8e = 142 bytes. JSON-RPC error object; the request ID as65001-4.4.4.4:35 ties it to the denied submit. Note what's absent: no capability leak — the error names only the skill and peer, not what is granted.

- NCFED header — length (uint32) + flags
- JSON-RPC 2.0 payload (UTF-8)

14:24:43.593893 · Nick → John — header 00 00 0c 25 00: length 0x0c25 = 3,109 bytes, flags 0 (no chunking needed). Carries state: "completed" plus the full markdown fleet-health report as output text. Readable in cleartext — see finding 4.

**1 · Authorization enforcement verified on the wire — fail-closed, then open after grant**

## 2 · Task 83d1176e stalled for 19 minutes and refused cancellation

### 3 · endpoint\_update re-announce loop is 64% of all NCFED JSON-RPC traffic

#### 4 · The federation channel is cleartext — chat and task results readable on the wire

## 5 · Protocol correctness & transport health: excellent

### Delivered payload — what the delegation produced

**Fleet Health Check — “NetClaw True CCIE Enterprise 20-Node v2”** (task 9751b33a..., completed 14:24:43, frame 853)

- **14/14 devices reachable**, all protocols converged: 50/50 BGP sessions UP, 26/26 OSPF FULL adjacencies, both EIGRP AS10/AS20 neighbors UP.
- Zero CRC / input / output errors fleet-wide; CPU 0–1%, memory uniform at 17.3–17.6%.
- Overall ⚠ **WARNING from a single cause**: every device's uptime  $\approx 37$  min ( $< 24$  h threshold) — lab freshly started; expected.
- R1/R2 correctly show 9 BGP sessions each (iBGP route reflectors); R7/R8/R12/R13 OSPF=0 by design (EIGRP branch LANs, eBGP-only ISP nodes).

## Reproducing this analysis

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```
sudo tcpdump -i any -nn -s 0 'tcp port 20203' -w n2n-session.pcap # capture both mesh + federation channels
tshark -r n2n-session.pcap -q -z conv,tcp # find the streams
tshark -r n2n-session.pcap -Y 'tcp.len>0' -T fields -e frame.time_epoch -e tcp.stream -e tcp.payload # extract payloads
# then parse: 13 B handshake (magic "NCFED" + !I AS + router-id), else 5 B !IB header + JSON-RPC body; len=0 ⇒ heartbeat
```